

National Technical University of Ukraine
“Igor Sikorsky Kyiv Polytechnic Institute”
Faculty of Informatics and Computer Science
Department of Information Systems and Technologies

Lab work No5

Time Series in Pandas

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Tasks

Create a Python program that does task according to your number in the group list.

Make a report in doc-file. Report must contain:

— Title;

— Program code;

— Results (screenshots) of code execution.

Demonstrate the working program and answer questions about theoretical information and program working.

File **digital_currency.csv**.

1. Build open_USD price graph: a) for entire period; b) for year 2019; c) for December 2020; d) for October 2018 – July 2020; e) for the last two weeks of September 2020.

2. Find the average low_USD price a) for 2019; b) for each month; c) for every spring week of 2020.

My Full Code:

```

import pandas as pd
import matplotlib.pyplot as plt

# Read data from the CSV file
df = pd.read_csv('digital_currency.csv')

# 1a. Open USD price graph for the entire period
df['Unnamed: 0'] = pd.to_datetime(df['Unnamed: 0'])
plt.plot(df['Unnamed: 0'], df[['open_USD']])
plt.title('Open USD Price for the Entire Period')
plt.xlabel('Date')
plt.ylabel('Open USD Price')
plt.show()

# 1b. Open USD price graph for the year 2019
df_2019 = df[(df['Unnamed: 0'] >= '2019-01-01') & (df['Unnamed: 0'] <= '2019-12-31')]
plt.plot(df_2019['Unnamed: 0'], df_2019[['open_USD']])
plt.title('Open USD Price for the Year 2019')
plt.xlabel('Date')
plt.ylabel('Open USD Price')
plt.show()

# 1c. Open USD price graph for December 2020
df_dec_2020 = df[(df['Unnamed: 0'] >= '2020-12-01') & (df['Unnamed: 0'] <= '2020-12-31')]
plt.plot(df_dec_2020['Unnamed: 0'], df_dec_2020[['open_USD']])
plt.title('Open USD Price for December 2020')
plt.xlabel('Date')
plt.ylabel('Open USD Price')
plt.show()

```

```

# 1d. Open USD price graph for October 2018 - July 2020
df_oct_2018_jul_2020 = df[(df['Unnamed: 0'] >= '2018-10-01') & (df['Unnamed: 0'] <= '2020-07-31')]
plt.plot(df_oct_2018_jul_2020['Unnamed: 0'], df_oct_2018_jul_2020[['open_USD']])
plt.title('Open USD Price for October 2018 - July 2020')
plt.xlabel('Date')
plt.ylabel('Open USD Price')
plt.show()

# 1e. Open USD price graph for the last two weeks of September 2020
df_sep_2020_last_two_weeks = df[(df['Unnamed: 0'] >= '2020-09-17') & (df['Unnamed: 0'] <= '2020-09-30')]
plt.plot(df_sep_2020_last_two_weeks['Unnamed: 0'], df_sep_2020_last_two_weeks[['open_USD']])
plt.title('Open USD Price for the Last Two Weeks of September 2020')
plt.xlabel('Date')
plt.ylabel('Open USD Price')
plt.show()

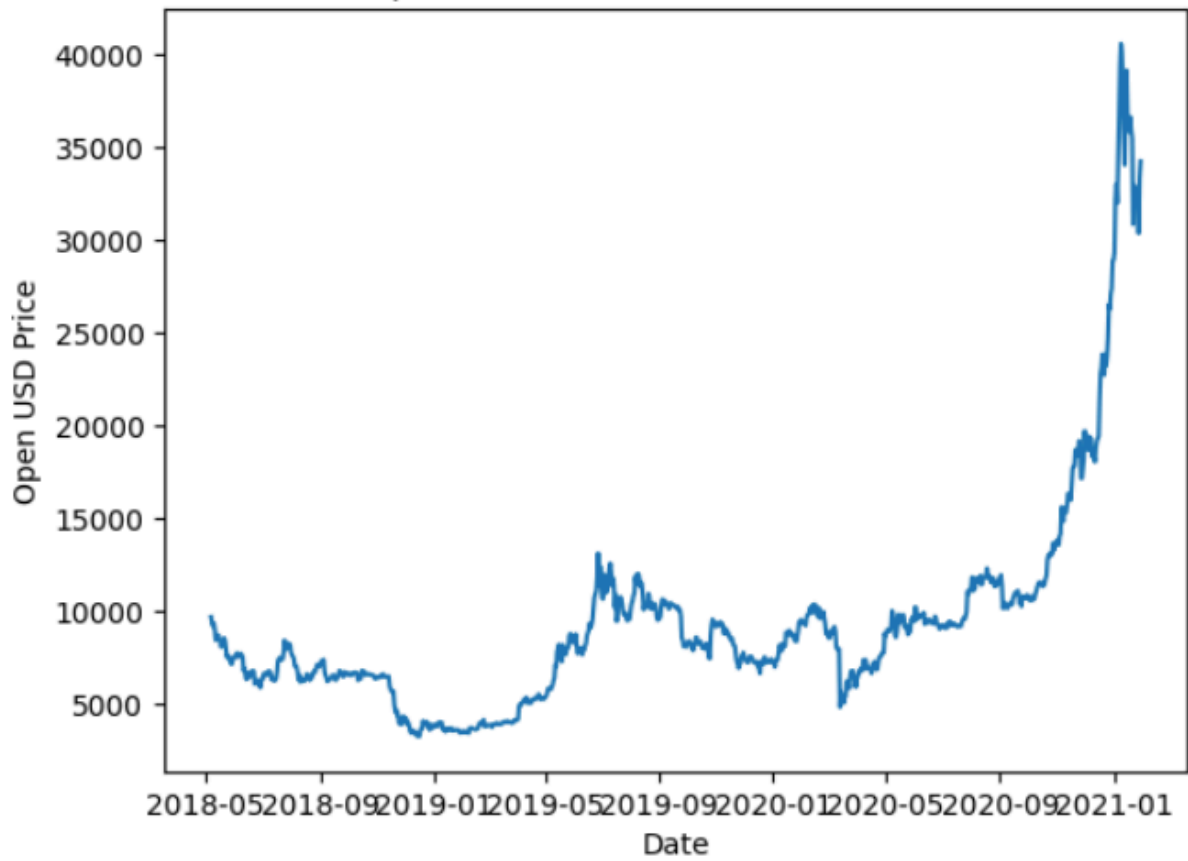
# 2a. Find the average low USD price for 2019
average_low_2019 = df_2019['low_USD'].mean()
print(f'Average Low USD Price for 2019: {average_low_2019}')

# 2b. Find the average low USD price for each month
df['month'] = df['Unnamed: 0'].dt.month
average_low_monthly = df.groupby('month')['low_USD'].mean()
print('Average Low USD Price for Each Month:')
print(average_low_monthly)

# 2c. Find the average low USD price for every spring week of 2020
df_spring_2020 = df[(df['Unnamed: 0'] >= '2020-03-01') & (df['Unnamed: 0'] <= '2020-05-31')]
average_low_spring_2020 = df_spring_2020.groupby(df_spring_2020['Unnamed: 0'].dt.isocalendar().week)['low_USD'].mean()
print('Average Low USD Price for Every Spring Week of 2020:')
print(average_low_spring_2020)

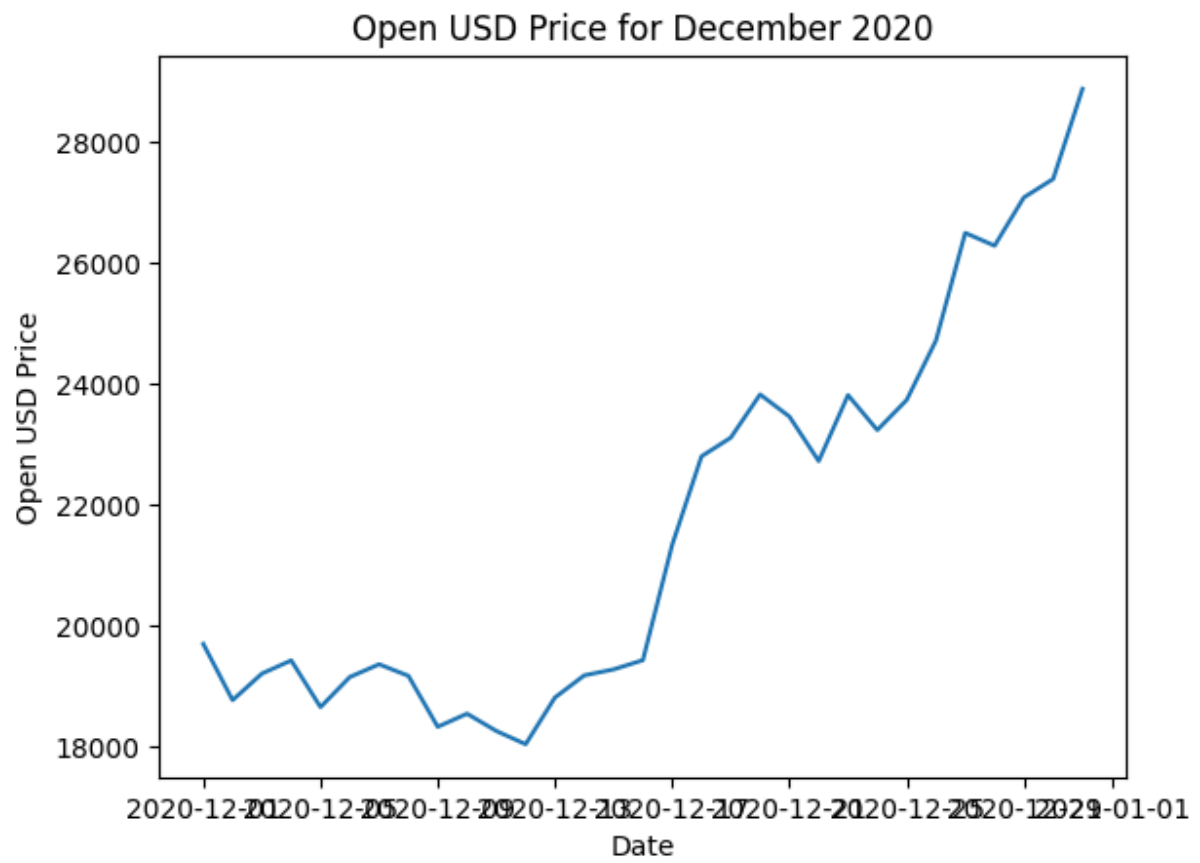
```

Open USD Price for the Entire Period

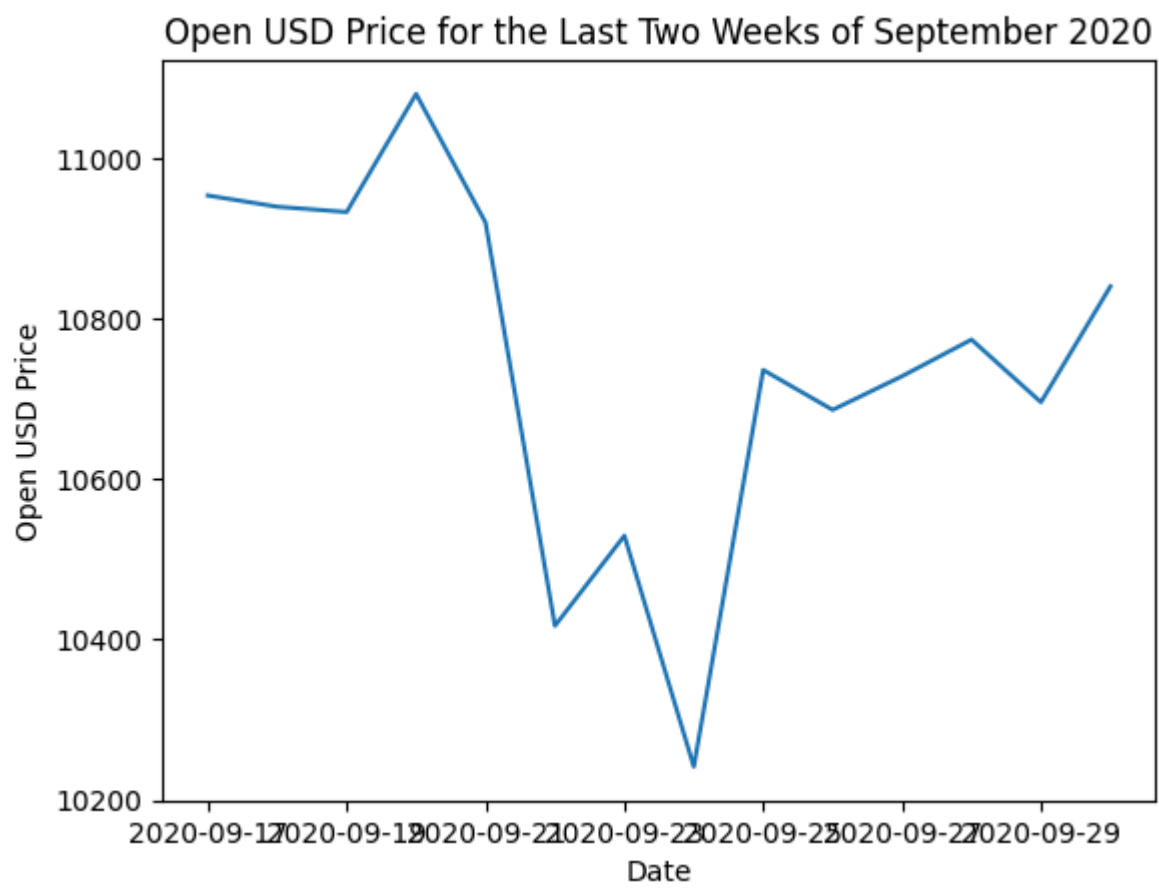


Open USD Price for the Year 2019









Average Low USD Price for 2019: 7152.498493150685

Average Low USD Price for Each Month:

month

1	14550.056739
2	6576.904561
3	5207.386452
4	5999.903167
5	7978.167356
6	8296.520222
7	8848.523656
8	9406.505269
9	8804.215556
10	8746.722796
11	9831.463889
12	10600.976667

Name: low_USD, dtype: float64

Average Low USD Price for Every Spring Week of 2020:

week

9	8411.000000
10	8625.148571
11	5898.625714
12	5269.928571
13	6167.270000
14	6401.414286
15	6911.437143
16	6777.932857
17	7092.867143
18	8207.818571
19	8935.887143
20	8924.375714
21	9111.821429
22	9043.870000

Name: low_USD, dtype: float64